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Comparing energy transitions in Germany and Spain using a political economy perspective

Tobias Haas

Forschungszentrum für Umweltpolitik (FFU) at Freie Universitaet Berlin, Otto-Suhr Institute for Political Science, Ihnestr. 22, 14195 Berlin, Germany

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ABSTRACT

During the 2000s, Germany and Spain were forerunners in Europe's energy transition. However, after the outbreak of the global financial crisis in 2007 a growing divergence in the two countries' energy policy paths can be observed. This article applies a political economy perspective to explore the reasons for this divergence. In investigating this empirical puzzle, the work identifies two main causes of the differing developments in the two countries. The first cause being that Spain was hit much harder by the financial and economic crisis and the second, that the Spanish energy transition was an elite-driven process whereas the German *Energiewende* was driven by large sectors of civil society. Regarding the theoretical question of how to understand the divergent developments of the countries' energy politics, a perspective based on regulation theory and Antonio Gramsci's political theory can help offer an understanding of the social character of energy transitions.

1. Introduction

In the 2000s, Germany and Spain were forerunners of Europe's energy transition. The energy transition's push for renewables is articulated in conflicts about society's relations with nature, set against the backdrop of the politicization of the ecological crisis, especially climate change. Together with Denmark, Germany and Spain experienced the fastest development of their renewable energy sector in Europe (Toke, 2011). The share of renewables in final electricity consumption grew in Germany from 6.6% in 2000 to 16.6% in 2010. At the same time in Spain, the share of renewables increased from 17.2% to 35.3%. However, within the last few years the development paths of the two countries have significantly diverged. While in Germany there is still a relatively stable growth path for renewables, in Spain the government has almost entirely suspended the country's energy transition. Retroactive cuts and sharp declines in remuneration for new facilities from 2008 onwards culminated in the so-called green moratorium that was adopted in January 2012, which suspended feed-in tariffs (FiT) for newly installed renewable facilities. Consequently the share of renewables in Spain's electricity sector only grew from 35.3% in 2010 to 37.4% in 2015, while Germany saw its share rise from 16.6% to 30.0% during the same period (Agora Energiewende, 2016a, REE, Red Electrica de España 2016).

There has been a wide variety of social science research focusing on different aspects of the *Energiewende* (Brunnengräber and Di Nucci, 2014; Gailing and Moss, 2016; Schill et al., 2016) and some research on the Spanish *transición energética* (Fabra Portela and Fabra Utray, 2012; del Río and Mir-Artigues, 2014d). Studies conducted by Bernd Hirschl (2008) on Germany's *Energiewende* and Micha Bechberger's (2009) work on renewable energies in Spain are particularly meticulous as they develop a profound understanding of the two countries' energy policies, with both works using the advocacy coalition framework approach to structure the

Abbreviations: (NGOs), Non-Governmental Organizations; (EU), European Union; (FITs), feed-in tariffs; (PV), photovoltaic

E-mail address: tobias.haas@fu-berlin.de.

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Table 1
Germany and Spain in Comparison.
Source: Own presentation.

	Germany	Spain
Economy	productive, active extraverted regime of accumulation, stable context factor	financialized, passive extraverted regime of accumulation, erratic context factor
(Civil-) Society	active consent for renewables active dissent against fossil and nuclear technologies	passive consent for renewables passive dissent against fossil and nuclear technologies
State	strong legislative power, deliberation, orientation on compromises, high responsiveness	strong executive power, intransparency, limited orientation on compromises, low responsiveness

field of conflict (Table 1).

Aside from these works, the field of sustainability transition studies has been gradually emerging over the last 20 years, based on the approaches of transition management, strategic niche management, the multi-level perspective and technological innovation systems (Markard et al., 2012: 956–959, Loorbach et al., 2017). However, there are several shortcomings in this line of research, especially as the works tend to follow a rather unpolitical understanding of transitions (Moss and Gailing, 2016) and lack a profound theoretical basis in the social conditions that constitute the context of sustainability transitions (Geels, 2011). I will further elaborate on this issue in the next section. In order to address these shortcomings, I will develop a comparison of Germany's and Spain's energy transition on the basis of a critical political economy perspective. This analytical perspective allows the identification of important determinants for the two countries' diverging development paths through an analysis of the conflicts and power struggles within the energy transition (Geels, 2010: 501–502) and how it is articulated within overall economic development, civil society and the state. The aim is to develop a sophisticated understanding of the main aspects of the energy transitions rather than generating new insights into market regulations, renewable support schemes or the spatial dimension of energy transitions. Furthermore, I focus on economic and political developments, knowing that technical and environmental factors also influence energy markets (e. g. the demand for electricity or the volume of feed-in of renewable energies like wind or solar).

The article is structured as follows: in the next section, a brief overview of transition debates will be given and central shortcomings will be examined. Building on these shortcomings, I will develop a critical political economy perspective based on insights from regulation theory and Antonio Gramsci's understandings of politics. This perspective will help illuminate the main reasons for the diverging development paths in Germany and Spain which are rooted in the economy, civil society and the state and its complex articulations. In the third section, I will discuss the *Energiewende* in Germany by focusing on the underlying political economy and the role of civil society and the state. In the fourth section, I analyze the Spanish energy transition and in the following section, I compare the two case studies. I then argue that the increased stability of the German *Energiewende* is based on a steadier economic environment, stronger support within civil society and a more compromise-oriented energy policy regulation that is reflected in the materiality of the state and the procedural character of decision-making within the state. I will finish with a conclusion and a forecast that is informed by the antecedent analysis.

In addition to the analysis of scientific literature, position papers and press releases by important actors, in 2013 and 2014 I conducted 45 expert interviews in Germany (22) and Spain (23) with a wide range of stakeholders (including politicians and representatives from ministries, NGOs, business associations, labor unions and think tanks). These interviews offered important insights into the power struggles over the energy transition in both countries. The interviews were supported by a set of guidelines and aimed to identify the strategic actions of the different stakeholders in their struggle over hegemony in German and Spanish energy politics. The questions covered the interests, ideological orientations and visions of the countries' future electricity regimes, as well as discursive strategies and political alliances employed in concrete struggles. The interviews were coded and transcribed according to Gläser and Laudel's (2009) method of qualitative content analysis and referring to the above-mentioned research question. Furthermore, the interview data was triangulated with primary and secondary sources to ensure its reliability. Interpreting the data collected allowed me to develop a sound understanding of the energy-related struggles and their links to the countries' political economies. I make reference to several interviews later in the work, especially in the sections on Spain as there exists only very limited literature on the political struggles over the energy transition in this country. The focus of this paper lies on developments up until 2015, with more recent events only intimated. Furthermore, the paper only analyzes the shift to renewable energies in the electricity sector and focuses on the renewables' share in total electricity consumption.

2. Transition theories and the political economy of green transformations

2.1. Transition theories

Within the last two decades, the academic debate on sustainability transitions has intensified remarkably. Markard et al. (2012: 955) distinguish four different leading theoretical frameworks: transition management, strategic niche management, the multi-level perspective on socio-technical transitions and technological innovation systems. Coming from different theoretical and disciplinary backgrounds, these approaches question the interplay between social and technical developments that cause a transition to sustainability. Sustainability transitions are understood as "long-term, multi-dimensional, and fundamental transformation processes

through which established socio-technical systems shift to more sustainable modes of production and consumption "(Markard et al., 2012: 956).

Research on sustainability focuses on different policy fields, but energy policy enjoys a very high level of recognition, as the decarbonization of the energy system is key to curtailing climate change. While most research takes the form of single case studies (Markard et al., 2012: 961), Geels et al. (2016) compare the electricity transitions in Germany and the UK by developing a dynamic understanding of different transition pathways. Defining different phases and analyzing the strategies, interests and institutional configurations of the political systems the authors show, that in the German *Energiewende* new actors play a more important role than in the UK's energy transition. The different transition pathways also stem from the countries' support schemes – the German FiT offered far greater opportunities for new actors compared with the UK's quota obligations. While this comparison offers important insights into the two countries' developments, later in the work I will propose an analytical framework that puts more emphasis on the economic and social developments that build the contextual conditions for the power struggles over energy transitions.

Against the backdrop of the lack of a profound understanding of contested social relations (Brand, 2016), sometimes disputable assessments of the current state of sustainability transitions occur. For example, Loorbach et al. (2017: 602) state that the "Sustainable Development Goals and the COP21 Paris Agreement signal that the need for change is no longer questioned, and the overall direction away from a fossil-based economy is clear." In the age of Trump and the rise of right-wing parties and movements across Europe which are often skeptical of the climate agenda, the abandoning of Germany's 2020 climate target and the enormous divergence between the nationally indicated action plans and the need for carbon reductions to meet the Paris Agreement, indicate that a profound understanding of the obstacles and barriers to sustainability transitions is necessary (Brand, 2016). Sterling argues that "it often remains rather non-specific and ambiguous what exactly will constitute these widely mooted 'green transformations' or 'transitions to sustainability'" (Stirling, 2015: 55).

Furthermore, transition theories tend to take a "broadly apolitical approach to socio-technical change" (Moss and Gailing, 2016: 6) and pose issues when conceptualizing the interplay between structure and agency (Svensson and Nikoleris, 2018; Hess and McKane, 2017). In light of this, several (self-)criticisms of transition theory have been developed regarding the ontological, the epistemological and the normative dimension (Geels, 2011). To overcome some of these shortcomings Geels (2014: 21) suggests a greater focus on incumbent regimes in order to bring "politics and power into the multi-level perspective." Instead, I develop a political economy approach that "provides the sort of understanding of power and structure that is absent in much of conventional theorizing about transitions and green transformations" (Newell, 2015: 70). I will do so by referring to regulation theory and Gramscian approaches in combination with critical currents in transformation studies.

2.2. The political economy of transitions

In the late 1970s French economists, influenced by indicative planners and Marxist theory, founded regulation theory. Against the backdrop of the long-term stability of the Fordist social formation, their key epistemic interest was to explore how capitalist societies are reproduced due to inherent contradictory relations and crisis tendencies. To generate answers to this question, regulation theorists developed analytical instruments that are located on an intermediary level of abstraction. These instruments allow for the analysis of historically and spatially concrete capitalist formations. In regulation theory the economy is, unlike the neoclassical concept, not a separate sphere but mutually interwoven and co-constituted with society and politics. To capture this nexus, regulation theory builds on two central concepts, the 'regime of accumulation' and the 'mode of regulation' (Bieling et al., 2016).

Alain Lipietz, one of the founding fathers of the theory, defines a 'regime of accumulation' as a "mode of systematic distribution and reallocation of the social product which, over a prolonged period, is able to coordinate transformations in the conditions of production…with transformations in the conditions of final consumption" (Lipietz, 1988: 23). While the 'regime of accumulation' refers to specific conditions necessary for the valorization of capital, the 'mode of regulation' refers to the embeddedness of the economy in a "totality of institutional forms, networks, and explicit or implicit norms assuring the compatibility of behaviors within the framework of a regime of accumulation" (Lipietz, 1988: 25).

Regarding the process of capital accumulation, Joachim Becker (2002: 67) distinguishes three different axes: first, productive vs. financialized accumulation, second, extensive vs. intensive accumulation, and third, introverted vs. extraverted accumulation. Due to the epistemic focus of this article, which is based on a comparison of the energy transition dynamics in Germany and Spain, I will only deal with the first and the third axis. Productive accumulation is performed if the valorization of capital is mainly based on investments in the productive sphere of the economy (industry, agriculture etc.) whereas financialized accumulation is based on expanding banking and financial markets, which often coincides with asset price bubbles. Regarding the third axis, introverted accumulation refers to a strong orientation toward domestic markets whereas extraverted accumulation is based on exports and current account surpluses (active extraversion) or capital and commodity imports that correspond with current account deficits (passive extraversion). As will be shown in the following chapters, the German model of capitalism features a productive, active extraverted regime of accumulation whereas Spain features a financialized, passive extraverted regime of accumulation (Becker and Jäger, 2012: 177–179).

Regulation theory is a helpful starting point for analyzing the political economy of energy transitions within a country as it allows us to capture analytically basic features of the models of capitalism and the co-constitution of the economy and the political sphere (the social and political embeddedness of markets). However, several problems remain with regulation theory, especially the difficulty to grasp the political contestations that are constituents of the regime of accumulation as well as the mode of regulation:

"regulation theory [.] runs the risk of suffering from an implicit functionalism, as the mode of regulation is primarily observed

through the way in which it stabilises the accumulation regime. This tendency is admittedly much more apparent when individual research adopts a socio-technological and regulatory problem-solving perspective that barely asks how societal contradictions and crises escalate and articulate specific power structures and conflicts." (Bieling, 2014: 36)

Regarding the application of the theory, several efforts have been undertaken to further develop regulation theory to capture the relations between society and nature within a regulationist framework (Görg, 2003; Lipietz, 2013). Connected to this issue, but still outstanding, is the question of how to apply insights from regulation theory to energy transitions. To fill this gap I will build on Antonio Gramsci's theory of politics and critical approaches to green transformations (Scoones et al., 2015).

Antonio Gramsci's understanding of politics exhibits many similarities with regulation theory, in particular regarding the ontology of the economy, civil society and the state, as well as the epistemic focus on the national level. However, Gramsci's thinking shows a greater sensitivity toward political struggles than regulation theory does. Influenced by his Marxist background, Gramsci contemplated in his prison notebooks on, among other issues, the question of how bourgeois authority is secured and renewed within capitalist societies. He squeezed his key considerations on this question into the term "hegemony". While Marxist scholars during his lifetime focused on the oppression of the workers, Gramsci shifted the perspective of his analysis by asking what makes the subaltern accept their own marginalization. This consideration paves the way for an analysis of moments of ideological leadership, common sense and everyday practices. Therefore, bourgeois hegemony is not only based on coercion but also contains the consent of the subaltern, which can only be achieved by material and ideological involvement in the context of power. In light of these considerations, political struggle in a Gramscian sense is about universalizing particular interests by representing them as the common interest (materially and ideologically) (Gramsci, 1992; Haas, 2019: 67–68).

The terrain of hegemonic contestation is the integral state, which compromises civil society and the state in a narrower sense. While Gramsci's differentiation between civil society and the state is not entirely coherent in his writings (Anderson, 1976), the following quotation gives an idea of his perception of civil society and the state:

"'civil society', that is the ensemble of organisms commonly called 'private', and that of 'political society' or 'the State'. These two levels correspond on the one hand to the function of 'hegemony' which the dominant group exercises throughout society and on the other hand to that of 'direct domination' or command exercised through the State and 'juridical' government." (Gramsci 1992: 12)

On other occasions, Gramsci concedes that the differentiation between civil society and the state is purely analytical and that in both spheres moments of force are articulated with consent. The theorization of the state was further developed by the materialist theory of the state. Building on Nicos Poulantzas (1978: 128–129) the state can be understood as a social relation, "a relationship of forces, or more precisely the material condensation of such a relationship among classes and class factions, such as this is expressed in the state in a necessarily specific form". Following this consideration, the state is not entirely separate from civil society but has a relative autonomy and its own materiality through its apparatuses. This understanding of the state as a conflictual materialization of social relations contrasts with liberal understandings of the state.

Regarding the involvement of the subaltern, Gramsci distinguishes on several occasions between spontaneous and active consent. While spontaneous consent is based on situational circumstances, active consent builds on the ability of the ruling class to involve the subaltern in a way that leads them to not only uphold, but also actively affirm the societal conditions that cause their own subordination (Gramsci 1992: 266, 370). This distinction is helpful in understanding the character of the people and civil society's involvement in the transition to renewable energy. At this point, we can use Stirling's work to distinguish between two forms of changes to renewable energies: transitions and transformations. While the first concept stands for a process that is driven by the elite, preserves the existing power relations and leaves society with a passive role, the latter stands for a process that follows a bottom-up logic with an active society and tends to less hierarchical power relations:

"Transitions […] are managed under orderly control, through incumbent structures according to tightly disciplined technical knowledges and innovations, towards a particular known (presumptively shared) end. This typically emphasizes integrated multidisciplinary science directed at processes of instrumental management trough formal procedures in hierarchical organizations sponsored by the convening power of government. Transformations, on the other hand, involve more diverse, emergent and unruly political alignments, challenging incumbent structures, subject to incommensurable, tacit and embodied social knowledges and innovations pursuing contending (even unknown) ends. Here there is a much stronger role for subaltern interests, social movements and civil society, conditioning in ambiguous and less visible ways the broader normative and cultural climates in which more explicitly structured procedures are set." (Stirling, 2015: 62)

Summing up these considerations and touching upon the comparative study on energy transitions that follows, we can borrow the insight from regulation theory that energy transitions are in articulation with a specific regime of accumulation that constitutes the contextual conditions regarding the demand for electricity, the financial resources available for a transition and so on. This regime of accumulation is stabilized by a specific mode of regulation that is better understood with reference to Gramsci's less functionalist understanding of politics. The process of changing energy systems can be understood as struggles over hegemony where different actors pursue specific aims and try to universalize their particular interests. These struggles embrace material interests (utilization of the existing energy infrastructures versus developing new spheres of accumulation) as well as ideological convictions (competitiveness, sustainability or security of the energy supply, as well as considerations on the ownership structure and democratic potential of the energy system).

3. The political economy of Germany's Enegiewende

The following section discusses the main characteristics of the German *Energiewende* focusing on the economic context and its anchorage within civil society and the state. The combination of a relatively stable regime of accumulation, active consent for the *Energiewende* within civil society and a comparatively responsive state led Germany's renewable energy transition to exhibit several characteristics of a green transformation as outlined above.

3.1. A stable economic context

Germany's economy has for a very long period shown a specialization in industrial processes, especially in the chemical, engineering and automotive industries. This industrial make-up leads to a strong outward orientation and a focus on making new markets accessible. The European integration process and the establishment of the Euro as a single currency, which strengthened the competitiveness of the German industry due to the relative undervaluation of the Euro, reinforced the profile and path dependency of the German economy. Despite not being immune to internal and external crises tendencies and business cycles, the German mode of capitalism is historically comparatively stable and in the current European crisis is outperforming its neighbors (Streeck, 2013). Due to the German economy's broad industrial base which secures high current account surpluses, the German regime of accumulation can be characterized as productive and active extraverted (Becker and Jäger, 2012: 178–179). In terms of the country's energy system, the comparatively stable economic growth rates offer a reliable framework. Demand for electricity is quite stable as efficiency gains enable economic growth without creating additional demand for electricity (Agora Energiewende, 2016a: 11-12). Between 2000 and 2015 total gross electricity consumption rose from by around 3% in Germany (Agora Energiewende, 2018: 23).

Furthermore, the *Energiewende* has the potential to renew German's industrial supremacy as German industry is heavily involved in the construction and development of renewable energy technologies. The construction and operation of renewable energy plants offers huge export potential for German industry due to the global boom in demand for renewable technologies. The "Export Initiative Renewable Energies" (Exportinitiative Erneuerbare Energien) was established by the German Ministry for Economy back in 2003 to help German enterprises access new markets (Haas, 2017: 172). The growing global demand for renewable energies has also had a significant impact on employment levels. In 2012, around 400,000 people worked in the renewable energy branch. However, due to the collapse of the German solar cell industry, the sector lost more than 40,000 jobs in 2013. The total employment figure in the renewables sector decreased from 399,800 in 2012 to 371,400 in 2013 (BMWi, 2015: 3). This indicates that the German economy is vulnerable to shifts and partial crises. But as the production of solar cells was of marginal importance compared to the chemical, engineering and automotive industries, neither the association of German industry (Personal communication BDI (Bundesverband der Deutschen Industrie), 2014 nor the German government pushed for trade measures to protect the domestic producers (Personal communication BDI (Bundesverband der Deutschen Industrie), 2014). In spite of the crisis that afflicted parts of the German solar industry, overall economic development in Germany has remained very stable (Haas, 2017: 145–155).

3.2. Active consent within civil society

Germany's *Energiewende* was pushed by the active involvement of (civil) society. The number of citizen energy cooperatives grew from 77 in 2005 to 754 in 2012 (trend:research and Leuphana Universität Lüneburg, 2013: 60). Many farmers invested in different forms of renewable energy technologies (especially in biomass and solar facilities), as did municipal utilities. The *Energiewende* offered the potential for municipal value creation in particular for rural areas. This strong anchorage within civil society resulted in "new actor landscapes" (Becker et al., 2012) and a strong push toward a decentralization of the energy system which was originally based on large fossil and nuclear power plants (Gailing and Röhring, 2016). In 2012, more than 40% of Germany's renewable energy came from citizen energy facilities while the share of traditional energy companies, the big four and municipal companies, was only slightly above 20% (trend:research and Leuphana Universität Löneburg, 2013: 46).

This indicates that German (civil) society is actively involved in the *Energiewende*. The path to a renewable energy regime follows in large part the logic of the above-defined transformation path. The strong involvement and active consent of civil society (Gramsci, 1992: 224, 370) in Germany dates back to the major ecological movements that emerged in the 1970s (Sander, 2016). These movements not only pushed for a phase-out of nuclear energy but also sowed the seed for the development of renewable energies (Toke, 2011: 71–72). Parallel to the development and growth of the renewables sector, green capital associations like the BEE (Bundesverband Erneuerbare Energien), BWE (Bundesverband Windenergie) or BSW (Bundesverband Solarwirtschaft) were founded in the 1980s and 1990s. They share a clearly antagonistic stance toward the old fossil-nuclear energy regime and its leading actors (Haas and Sander, 2013: 13–17). Within the last few years and especially since the Fukushima nuclear accident in Japan, initiatives supporting the phasing-out of coal have gained momentum with a broad variety of actors from NGOs to social movements pushing for the fast-tracking of a renewable energy system (Haas and Sander, 2016).

The active consent for renewables in Germany is connected with an active dissent against nuclear and fossil technologies. This nexus indicates that the resistance of incumbent regimes is not a flipside of energy transitions. Rather the struggle to achieve a renewable energy regime is in steady articulation with the struggles of the actors driving the old energy system (Paul, 2018).

The main instrument to promote the *Energiewende* is the Renewable Energy Act (Erneuerbare Energien Gesetz, EEG), which was passed by the German parliament in 2000. The law guarantees priority feed-in for renewable energies and a fixed remuneration for each kilowatt-hour produced. The initiative for this law came from a number of parliamentarians, who pushed for the law despite the resistance of the leading ministry of economics (Hirschl, 2008: 59–196). This indicates that the forces pushing for a fast transition to

an energy regime based on renewables succeeded in inscribing their interests in state policies (Sander, 2016: 123–133). However, we should not fail to mention that after Fukushima social forces opposing a fast and decentralized Energiewende fiercely attacked the EEG, blaming renewables for being too expensive and threatening the competitiveness of German industry (Gawel et al., 2013).

3.3. A responsive state

Responsibility for energy policy within the German state apparatuses was divided in the period between 2002 and 2013 between two ministries: the ministry of the environment (BMU) and the ministry of economics (Personal communication BMWi (Bundesministerium für Wirtschaft), 2014). In 2000 the EEG was introduced as a parliamentary initiative in the face of resistance from the BMWi. From the outset, German electricity-consuming industries enjoyed widening exemptions from the levy to protect the productive, active extraverted regime of accumulation and leaving the main financial burden on private consumers and non-exempted businesses (arepo consult, 2012). After the introduction of the EEG in 2000 against the resistance of the BMWi, the government delegated responsibility for the EEG to the BMU in 2002. Traditionally this ministry has had a greater affinity with renewable energies while the BMWi has close ties with the fossil-nuclear energy regime and its principal actors (Hirschl, 2008: 452). Although most of the other political competencies remained in the hands of the BMWi, this shift allowed the different governments, also against the backdrop of the German parliament's strong position within the law-making process, to pursue an energy policy that took strong consideration of its impacts on the climate. From 2002 onwards, the struggle over the EEG within civil society was represented to a large extent within the state apparatuses, with the BMWi showing skepticism towards the *Energiewende* and the BMU supporting it (Sander, 2016: 107–112).

Due to the inclusion of the federal governments and the federal assembly, the energy legislation process in Germany is on the whole comparatively transparent and oriented towards finding a broad compromise (Personal communication BMWi (Bundesministerium für Wirtschaft), 2014). The *Energiewende* was driven by relevant parts of the state because of civil society's active consent for the process and the intensive debate about climate change that resulted in binding renewables targets at the European level and greenhouse gas reduction targets enshrined in the Kyoto protocol and European legislation. In spite of ongoing attacks on the EEG, it was not abolished (Gawel et al., 2013). Rather in 2004, the German government together with the Spanish government established the International Feed-in Cooperation (IFIC) as an expert forum for deliberations on feed-in systems. In addition the IFIC also became an important space for coordinating the defense of the feasibility of feed-in tariffs in the context of the EU renewable energy guideline from 2009 (Futterlieb and Mohns, 2009). This indicates, that renewable energy legislation is articulated within the process of European integration and that efforts were needed to defend the EEG. Due to the successful defense of the EEG, the *Energiewende* had a strong legal backing and a stable legal basis to develop further (Haas, 2017: 212–216).

However, in 2013 the responsibility for the EEG shifted back to the BMWi, 2015. This changing composition of the state mirrors the partial loss of terrain incurred by the social forces pushing for a renewable and decentralized energy regime (Paul, 2018: 3–7). The EEG amendment in 2014 (and in 2016) initiated the major move to abandon FiTs. Critics of renewable energies focused their critique on the increasing EEG levy, narrowing the discourse from total costs of electricity and long-term benefits of renewable energies to short-term consumer costs (Lauber and Jacobsson, 2016: 154–158). These developments symbolize a partial loss of hegemony for the *Energiewende* (Haas, 2017: 212–216).

4. The political economy of Spain's *transición energética*

The following section discusses the main characteristics of the Spanish *transición energética*. The aim is to analyze the articulation of the economic context in social and energy-related struggles within the 'integral state' (civil society and the state in a narrower sense). The Spanish journey to renewable energies was largely driven by the elite, based purely on passive consent and contains the basic logic of a transition as outlined in Section 2.2.

4.1. An erratic economic context

The Spanish economy has a very different profile to that of Germany. At the beginning of Franco's dictatorship, Spanish authorities geared economic policy towards autarky. From the late 1950s onwards, the Spanish government took the first steps towards economic liberalization leading to the emergence of industrial conglomerates. Nevertheless, the economy was still largely based on agriculture with tourism and construction growing in importance. After the transition to democracy in the late 1970s, the Spanish economy was gradually further integrated into the European economy. In 1986, Spain became a member of the EC. The Spanish government directed the economic, fiscal and monetary policy in line with European requirements: liberalization, privatization, price stability and austerity. However, large parts of Spain's industry suffered from poor competitiveness due to low productivity. This caused the specialization on tourism and the construction sector to deepen. Thanks to remittances from the diaspora, the current account deficit could be financed alongside a growth in foreign debt (López Hernández and Rodríguez, 2011).

From 1995–2007, Spain experienced a large-scale economic boom that pushed the GDP per capita close to the European average. Massive capital imports fueled the boom but coincided with high private levels of debt. Creditors invested large amounts in real estate, causing an asset price bubble. Due to the concurrence of the bursting of the real estate bubble and the spread of the world financial and economic crisis, the Spanish economy experienced a recession between 2008 until 2014 (Royo, 2015). With its the weak industrial base, the high dependence on capital imports and the strategic importance of the banking system, the Spanish regime of accumulation can be characterized as financialized and passive extraverted (Becker and Jäger, 2012: 177–178).

These cyclical developments are largely mirrored in Spain's energy policy and the transition dynamics toward renewables. During the economic boom, which was fueled by the electricity-consuming construction sector, annual average growth in electricity demand was around 5% (Bechberger, 2009: 669). This additional demand was mainly fed by new wind power and gas power stations and to a lesser degree by new photovoltaic and solar thermoelectric power stations. As planning horizons span several years and the crisis was unexpected, Spain now faces massive overcapacity in its electricity generation portfolio (Personal communication CNMC (Comisión Nacional de los Mercados y la Competencia), 2014; Personal communication Endesa, 2014; and Personal communication Iberdrola, 2014). Between 2008 and 2015 gross electricity consumption decreased by around 5% (REE, Red Electrica de España, 2016). The largely unused and not yet fully amortized gas power stations exacerbate the future distributional conflicts, which will be fought out in the integral state (Personal communication AEE (Asociación Empresarial Eólica), 2014; Personal communication Iberdrola, 2014). The erratic development of the peripheral, financialized and passive extraverted regime of accumulation serves as a source of instability that severely threatens the *transición energética*.

4.2. Passive consent within civil society

The strong anti-nuclear movement and its search for alternatives has been a driver for Spain's *transición energética* since its inception. One of the world's first wind turbine constructors was the Catalan-based cooperative Ecotècnica, whose founders were active in the movement against nuclear energy. Parts of the state supported these early alternative energy initiatives, due to Spain's high dependence on energy imports. Renewables were considered as a way of decreasing external energy dependency (Puig and Boix, 2009) but the nuclear moratorium of 1984 and the growing de-politicization of Spanish society after the transition to democracy caused a downturn in the activities of social movements (Huke, 2017: 145–147). Spanish society assumed an increasingly passive role in the *transición energética*. Corporations and parts of the state drove the energy transition (Personal communication, FR). Due to the speed of technological developments, suitable geographical conditions and an ever-growing demand for electricity during the economic boom, the ground was prepared for the rapid development of renewable energies (Bechberger, 2009). A hegemonic constellation ensured in particular the fast development of wind energy. However, due to the low involvement of society and conflicts of lower intensity, this constellation was grounded only in passive consent (Gramsci, 1992: 224, 370).

While new wind and gas power plants satisfied the increasing energy demand, the big energy corporations could continue to run their traditional business. They were also included in the growth constellation as they made huge investments in gas and wind power facilities. Iberdrola became the world's largest wind park operator (Galán, 2013). Along the renewables' value chain, several small and medium-sized companies developed their business models and articulated their interests via business organizations such as APPA (Personal communication APPA (Asociación de Empresas de Energías Renovables), 2013), AEE (Personal communication AEE (Asociación Empresarial Eólica), 2014) or Personal communication UNEF (Unión Española Fotovoltaica), 2014. In contrast to wind energy, the big energy companies did not participate in the short-lived boom in photovoltaics in 2007 and 2008 that was driven by excessive remunerations outlined in the Royal Decree 661/2007 (RD 661/2007) (del Río and Mir-Artigues, 2014d). Despite big energy companies partially participating in the 2009 and 2010 boom in solar thermoelectric installations, together with think tanks like FAES (Fundación para el Análisis y los Estudios Sociales), which is close to the conservative party PP (Partido Popular), these big companies started to take a more negative stance towards renewable energies (Personal communication FR (Fundación Renovables), 2014; Personal communication Px1NME (Plataforma por un Nuevo Modelo Energético), 2014). Solar energy in particular was the subject of fierce attack. Renewable support schemes were blamed for the escalating tariff deficit.

"Tariff deficits are shortfalls of revenues in the electricity system, which arise when the tariffs for the regulated components of the retail electricity price are set below the corresponding costs borne by the energy companies." (Johannesson Lindén et al., 2014: 3) In Spain price regulation was established under the electricity market law of 1997. Several factors aside from the generation of electricity from established technologies caused additional costs, such as renewable generation incentives, support for domestic coal, compensation related to the nuclear moratorium, capacity payments, transmission and distribution and the additional costs for the non-peninsular territories etc. In 2012, the tariff deficit totaled nearly 22 billion euros. This equated to 2 per cent of Spanish GDP (Paz Espinosa, 2013b). While the amount of the tariff deficit is indisputable, the complexity of its composition leaves scope for interpretation. My interview partners presented very diverse explanations, some arguing in a similar vein to the authors of a FAES study, which concludes that the electricity tariff deficit "has its fundamental origin in the uncontrolled growth in the volume of the remunerations for renewable energies" (Navarrete and Mielgo, 2011: 11) (Personal communication, CNMC, UNESA, Iberdrola, Personal communication UNESA (Asociación Española de la Industria Eléctrica), 2014). Others blame the large energy companies and their allies for instrumentalizing the tariff deficit to provide a reason for stalling the *transición energética* and protecting their fossil and nuclear assets (Personal communications APPA, FR, Greenpeace, P × 1NME and UNEF). The tariff deficit became, in accordance with the overall austerity-driven handling of the crisis, omnipresent in Spanish energy policy: "Any debate on energy regulation in Spain is currently dominated by the overwhelming problem of the so called 'electricity tariff deficit'" (del Guayo, 2015d: 354).

As the big energy companies were the main actors in the Spanish *transición energética* and turned their back on renewables after the outbreak of the crisis, the actors pushing for a renewable energy regime, mainly environmental NGOs and small companies and associated bodies such as APPA and UNEF, were far too weak to counter the attacks against renewable energies (Personal communication, UNEF). Although there is a broad rejection of nuclear energy in Spain (Foro Nuclear, 2013: 39) and some initiatives are fighting against fossil energy, these forces are not strong enough to ensure the country turns its back on the traditional energy regime. Passive consent for renewables is connected with passive dissent against the incumbent regime. Consequently, dramatic changes to federal legislation were made at the expense of renewables with retroactive cuts to funding for existing facilities and a suspension of feed-in tariffs for new ones (Paz Espinosa, 2013a).

However, we should also mention that there has been a politicization within Spain against the austerity policies. Emblematic for the political unrest is the May 15, 2011 (M15) movement when people all over Spain occupied squares. This coincided with demonstrations against cuts and privatizations in education (marea verde) and the health system (marea blanca) or against the forced eviction of people unable to pay their mortgage (PAH) (Huke, 2017). These activities are linked to an increasing politicization of the energy issue. The energy cooperative Somenergia and the think tank Renewables Foundation (Personal communication FR (Fundación Renovables), 2014) were founded in 2010 and 2012 saw the founding of the grassroots initiative Platform for a New Energy Model (Plataforma por un Nuevo Modelo Energético). These organizations demonstrate against the current energy model and the therein-inscribed power relations and demand the fast-tracking of a decentralized, renewable energy regime (Personal communication FR (Fundación Renovables), 2014; Personal communication Px1NME (Plataforma por un Nuevo Modelo Energético), 2014; Personal communication UNEF (Unión Española Fotovoltaica), 2014); (Capellán-Pérez et al. 2018).

4.3. A hardened ensemble of state apparatuses

Within the ensemble of the Spanish state apparatuses, responsibility for energy policy lies in the hands of the ministry of industry (MINETUR). While traditionally the senior ranks of the ministry are close to the huge energy companies, the subdivision for diversification and energy saving or IDEA (Instituto de Diversifiación y Ahorro de Energía), founded in 1984, tends to support renewable energies (Bechberger, 2009: 311). Although the Spanish parliament approved the 1997 and 2013 general electricity laws, the executive branch still has a very strong influence on Spanish energy policymaking. The general law has to be affirmed by Royal Decrees (Real Decretos, RDs) that are passed by the government and Royal Decree Laws (Real Decreto Leys, RDLs) that have to pass the parliament as well (Pause, 2012). But due to the executive branch's strong position (in accordance with the general composition of the Spanish policymaking process) the law-making processes lack transparency and deliberation (Huke, 2017: 161–164). There is not a strong need to find solutions that are mediated through compromises as the government's strong position allows it to push hard in one or another direction. These tendencies together with the weak involvement of Spanish society and, since the outbreak of the economic crisis, a rapidly changing economic environment have led to erratic policy changes that have harmed the Spanish energy transition (Pause, 2012).

During the economic boom, Spanish energy policy focused on securing additional demand by supporting the construction of new wind parks and gas power plants while keeping electricity prices low. These low prices were not only welcomed by private consumers but also fueled the construction sector, which is a major electricity consumer (Bechberger, 2009). At the climax of the boom, the RD 661/2007 was passed which guaranteed an excessive overcompensation of electricity from photovoltaic panels attracting Spanish as well as international investors (Sebastián, 2013; del Río and Mir-Artigues, 2014). When for several reasons the electricity tariff deficit rose, the parameters of energy policy regulation changed. Energy policy focused increasingly on the elimination of the tariff deficit in line with overall austerity policies. Consumers experienced sharp price increases and each operator suffered some cuts (Fabra Portela and Fabra Utray, 2012: 90). But the main burden within the electricity sector was put on the renewable energy operators (Paz Espinosa, 2013a). Large energy companies like ENDESA or IBERDROLA were able to stabilize their profits, benefiting from their high degree of internationalization (Personal communications, Endesa and Iberdrola). As there were several retroactive cuts for existing renewable facilities from 2010 onwards, there is huge legal uncertainty and the struggle shifted from the political to the legal terrain (Personal communication, APPA). Several lawsuits are pending in Spain, at the European level and at the World Bank's court of arbitration (Personal communication Holtrop (law office), 2014) A. This erratic policymaking and the legal insecurities further destabilize the Spanish *transición energética,* which has almost ground to a halt. However, as the Spanish economy has started to grow again, tenders being prepared and some municipalities are trying to push for the installation of renewables facilities, there is a window of opportunity to revive the Spanish *transición energética (*Akizu et al., 2017: 18056-18057*)*.

5. Comparing economic and political factors behind the German and the Spanish energy transition

Germany and Spain were both forerunners in Europe's electricity transition in the 2000s. The two federal governments established the IFIC as an expert forum to strengthen FiTs as the key cornerstone of accelerating the countries' move towards renewables (although recently there has been a partial shift to tenders in both countries). The two countries share several more commonalities, such as the emergence of quite important domestic renewable energy industries. Applying a combination of a regulation theory and a Gramscian perspective to Germany and Spain's energy transitions offers important findings on why fundamental differences between the two countries evolved within the last years.

While the German regime of accumulation is productive and active extraverted, the Spanish regime of accumulation is peripheral, financialized and passive extraverted. Although the two countries pertain to the EU and the Eurozone, fundamental inequalities between the core and the periphery persist, as the current economic crisis reveals (Becker and Jäger, 2012). The comparatively robust German economy delivers a much more stable context for the track to renewables than the Spanish economy does. This relates to the funding of renewable energy investments as well as for the demand for electricity, the latter of which is much more erratic in Spain (REE, Red Electrica de Espana 2016: 13–25, Bechberger, 2009: 669, Agora Energiewende, 2018: 23).

The expansion of renewables led to less conflict in Spain than it did in Germany because of the fast-growing demand for electricity. However, the comparatively high involvement of German (civil) society and strong social movements delivered active consent for the *Energiewende*, while Spanish society remained largely passive. This became obvious, when the economic crisis broke out and supporters of the energy transition within civil society, such as green business organizations and environmental NGOs, were too weak to defend the transition (Personal communication FR (Fundación Renovables), 2014; Personal communication UNEF (Unión

Española Fotovoltaica), 2014). The active involvement of German society was mirrored by strong political incentives to phase out nuclear energy and more recently by the intensification of efforts to eliminate coal (Agora Energiewende, 2016b). In Spain, there were also strong anti-nuclear movements but they were pacified by the moratorium of 1984 and as part of the more general pacification of Spanish society after the transition to democracy. However, in both countries the interest-based narrowing of the discourse on short-term consumer prices in Germany (Lauber and Jacobsson, 2016: 154–158) and on the Spanish tariff deficit paved the way for shifts in renewable energy policies in Germany and the stalling of the Spanish energy transition.

This passive role of Spanish civil society was enhanced by the make-up of the country's state and parliamentary system. While the executive branch has a very strong position with regards to policy making, the parliament's position is far weaker; political parties in Spain lack internal democracy and the election system is oriented on the two-party system (Huke, 2017: 150–154). These tendencies are mirrored in energy policy regulation which lacks transparency and deliberation processes and is therefore neither oriented towards compromise nor stability (Pause, 2012). The lack of reliability of the political system combined with the unsteady nature of economic development have made policy making on energy prone to erratic decision-making. On the other hand, Germany has a long corporatist tradition, established federal procedures and the legislative body enjoys a stronger position. These features tend to generate a higher degree of deliberation, broader compromises and a higher reliability of the policy outcome, which is further strengthened by the comparatively stable economic development (Haas, 2017: 277–286). Both energy transition pathways are deeply entrenched in their respective countries' political economy/regimes of accumulation and the historical character of the integral state as the site of hegemonic struggles in energy politics.

6. Conclusion

Applying a critical political economy perspective to the German and the Spanish energy transition has proved useful in understanding the modes of articulation between the economy, civil society and the state. The underlying focus on the dimension of power struggles offers insights into the social character of the two energy transitions and how different actors try to generate hegemony. This article contributes therefore to efforts to develop a more profound understanding of energy transitions by using social science theories (Gailing and Moss, 2016) and especially to the endeavor to bridge the gap between political economy and sustainability transition studies (van de Graaf, 2016). The comparison of Germany and Spain from a critical political economy perspective is of particular importance, as it allows us to explore why two forerunner countries took such fundamentally different directions in their energy policy between 2010 and 2015. While regulation theory helps us to understand the differences in the countries' political economy, Gramsci's understanding of hegemony and the integral state helps us to comprehend the dynamics of power struggle and its institutional enclosures. This analytical perspective also offers us a comprehensive understanding of transition processes and challenges traditional transition theories to place more emphasis on structural contexts and how these are articulated through agency and power struggles (Svensson and Nikoleris, 2018).

However, due to space restrictions I was only able to focus on the main features that characterize the energy transitions in Germany and Spain. Further research employing a critical political economy perspective on energy transitions could investigate at least four further aspects. First, the political economy of the spatial dimension of new energy infrastructures is of great importance. Developing a profound understanding of this aspect requires overcoming the bias towards the national level, as exploited in this study, as shifting energy infrastructures coincide with diverging material interests and dynamics within the multi-scalar state (Becker et al., 2016; Gailing, 2018).

Second, a more profound understanding of the interplay between economic context conditions and concrete policies would be fruitful. As is only briefly indicated in this paper, the support policies and the regulatory framework are important factors for each energy transition. There is a need for a better understanding of the nexus between actors' interests, strategic approaches and the struggles within the integral state. This is especially true, as renewable energy policies within the European context have experienced a quite radical shift away from FiTs towards tenders (Haas, 2017: 140–144).

Another, third aspect that requires further exploration is the aspect of ownership and how energy transitions do or do not fit in to changing regimes of accumulation. As there is a trend towards a greater financialization of national economies, the question of how far this also applies to the funding of energy-related investments, is a topic that could benefit from further research (Evans and Herr, 2016).

Fourth, greater emphasis could be given to (local) resistance to new energy infrastructures. This is especially important as, despite the energy transitions in Germany and Spain still enjoying support amongst the population, there are indications in Germany at least, that resistance is growing (Eichenauer et al., 2018). With the strengthening of the far-right party (AfD) which is skeptical of climate change and clearly opposed to renewable energies, it is important to understand the implications of the political shift to the right for the *Energiewende*. However, this is not only a German phenomenon. A critical political economy perspective can help us to better understand these developments and offer us insights into how it is articulated within energy policy.

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